Highlights

Satellite-Detectable Anthropogenic Fingerprints in Reservoir Storage Variability: Multi-Method Attribution Evidence from the Damodar Valley Corporation System Using Explainable AI

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- Sentinel-1 SAR reconstructs Maithon and Durgapur storage anomalies at 20 m resolution over 2017–2024, validated to RMSE = 0.07 m.
- XGBoost+SHAP isolates an anthropogenic residual (ΔS_{anthro}) of $[-0.19, +0.23]$ BCM (95 % CI ± 0.05 BCM) robust to lag specification.
- CCM with 999 phase-randomised surrogates confirms causal independence of ΔS_{anthro} from $\Delta S_{\text{climate}}$ and detects Maithon→Durgapur hydraulic coupling ($\varrho = 0.294 > \text{surr}_{95} = 0.144$).
- GRACE-FO partial regression yields $\beta_2 = +3.84\,\text{cm}\,\text{BCM}^{-1}$ ($p < 0.001$, 95 % CI $[+2.1, +5.6]$), consistent with theoretical basin sensitivity of $4.28\,\text{cm}\,\text{BCM}^{-1}$.
- DTW operational fingerprint taxonomy (silhouette = 0.49) classifies DVC reservoir operational regimes from satellite data alone, without in-situ sensors or discharge records.

Satellite-Detectable Anthropogenic Fingerprints in Reservoir Storage Variability: Multi-Method Attribution Evidence from the Damodar Valley Corporation System Using Explainable AI

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Abstract

Reservoir operations represent a dominant but poorly quantified source of terrestrial water storage (TWS) variability in heavily managed river basins. Basin-scale satellite analyses — including GRACE-FO TWS decompositions — routinely attribute this variability to climate-forced groundwater and soil moisture dynamics, potentially mischaracterising the true driver and misleading water stress assessments. This study develops a multi-method framework to isolate and characterise anthropogenic storage signals in the Damodar Valley Corporation (DVC) reservoir cascade, located in the Jharkhand–West Bengal border region of eastern India. Monthly surface water area was reconstructed for Maithon Dam and Durgapur Barrage over 2017–2024 from Sentinel-1 C-band SAR using adaptive Otsu thresholding constrained to the JRC permanent water mask, and converted to live storage anomalies via DEM-based area–volume hypsometry validated against nine independent CWC measurements (RMSE = 0.07 m, $n = 8$, excluding one temporal mismatch point). An XGBoost gradient boosting model trained on eight non-collinear climate drivers with expanding-window cross-validation predicted the climate-forced storage component (NSE = 0.208 on physically resolved rows; KGE = 0.226). The residual — the anthropogenic storage anomaly ΔS_{anthro} — ranged from −0.19 to +0.23 BCM (95 % CI width ±0.05 BCM, seed sensitivity $n = 50$), with a mean of −0.03 BCM indicating net management-driven drawdown. Climate features alone, without operational lag terms, yielded NSE ≈ 0.02, confirming that DVC operations — not climate variability — are the dominant driver of sub-annual storage fluctuations. Convergent cross-mapping (CCM) with 999 phase-randomised surrogates confirmed causal independence of ΔS_{anthro} from $\Delta S_{\text{climate}}$ ($\varrho = 0.044$ vs. $\text{surr}_{95} = 0.371$) and detected significant hydraulic connectivity from Maithon to Durgapur ($\varrho = 0.294 > \text{surr}_{95} = 0.144$, converging). A Hidden Markov Model identified two operational regimes: DRAWDOWN (mean = −0.168 BCM, 25 % of months) and FILL (+0.033 BCM, 75 %). Dynamic Time Warping clustering of annual storage cycles produced an operational fingerprint taxonomy (silhouette = 0.49) using satellite data alone, without in-situ sensors or discharge records. Partial regression against GRACE-FO RL06.3 V4 TWS anomalies yielded a significant coefficient for ΔS_{anthro} ($\beta_2 = +3.84\,\text{cm BCM}^{-1}$, HAC-corrected $p < 0.001$, 95 % CI $[+2.1, +5.6]\,\text{cm BCM}^{-1}$; partial $R^2 = 0.187$), consistent with the theoretical basin sensitivity of $4.28\,\text{cm BCM}^{-1}$. These findings demonstrate that DVC operations produce causally attributable storage signals detectable by multiple independent satellite methods, and that standard GRACE-based water stress assessments are likely to systematically misattribute a substantial fraction

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of Damodar TWS variability to climate forcing. The operational fingerprint framework offers a proof-of-concept framework for auditing reservoir-driven GRACE contamination across South Asian managed basins using freely available data.

Keywords: Reservoir operations, Anthropogenic attribution, Damodar Valley Corporation, Sentinel-1 SAR, GRACE-FO, Convergent cross-mapping, XGBoost, SHAP, Operational fingerprint taxonomy, India water management

1. Introduction

Terrestrial water storage (TWS) anomalies retrieved from the Gravity Recovery and Climate Experiment Follow-On (GRACE-FO) satellite mission have become the primary observational basis for diagnosing water stress at basin and continental scales (Tapley et al., 2019). In India, GRACE-derived depletion rates have shaped national groundwater policy, underpinned high-profile estimates of aquifer exhaustion in the Indo-Gangetic Plain (Rodell et al., 2009), and informed the first Indian National Water Policy framework's quantitative targets for sustainable withdrawals. Yet the attribution of GRACE TWS anomalies to specific processes — groundwater pumping, climate-forced soil moisture changes, or surface reservoir operations — is complicated by the coarse spatial resolution of GRACE mascon products ($\sim 90{,}000\,\mathrm{km}^2$ per cell), which integrates all storage changes across multiple basins and cannot resolve individual reservoirs, aquifer units, or irrigation districts. As a result, managed reservoirs — which store, release, and transfer water deliberately according to human operational mandates — are routinely conflated with the climate signal in GRACE decompositions (Wada et al., 2014).

The magnitude of this conflation problem is substantial. India operates over 5,000 large dams and more than 90 major reservoirs under central regulation, with a total live storage capacity exceeding 250 billion cubic metres (BCM). Seasonal pre-monsoon drawdowns (for flood preparedness) and post-monsoon conservation cycles (for irrigation delivery) produce TWS anomalies of -0.1 to $-0.5\,\mathrm{BCM}$ per reservoir per month — signals that are indistinguishable, at GRACE resolution, from equivalent groundwater depletion driven by agricultural pumping. If even a fraction of the GRACE-detected "groundwater depletion" in Indian river basins is attributable to reservoir operational cycles, current assessments of aquifer health and water security are systematically biased.

Progress on this problem requires two complementary advances: (1) a method to isolate the anthropogenic reservoir storage signal from the climate-forced background at reservoir-management resolution, and (2) causal evidence that the isolated signal is genuinely attributable to operations rather than to model underfit or unaccounted climate drivers. Neither advance has been demonstrated at the scale of an individual Indian reservoir system with multi-method cross-validation. Attribution studies at global scale (Ji et al., 2023; Wada et al., 2014) apply statistical decompositions to GRACE data directly, without independently resolving reservoir dynamics. Studies that do resolve individual reservoirs from satellite altimetry or optical imagery (Besseling et al., 2021) typically do not attempt causal attribution. And interpretable machine learning approaches that have transformed attribution in other Earth system contexts (Lundberg and Lee, 2017) have not yet been combined with causal inference frameworks for reservoir management applications.

The Damodar Valley Corporation (DVC) reservoir system in Jharkhand and West Bengal, eastern India, offers a well-constrained case study for addressing these gaps. The DVC operates three structures in a cascade: Maithon Dam (design capacity 1.360 BCM), Panchet Dam (0.290 BCM), and Durgapur Barrage (run-of-river), serving the dual mandate of flood control for the historically flood-prone Damodar floodplain and seasonal irrigation for approximately 400,000 hectares of Rabi cropland in West Bengal. The combination of a well-documented operational mandate, known historical flood events (notably the catastrophic 2009 flood; Maiti and Bhattacharya 2011), distinct reservoir types (storage vs. barrage), and a heavily instrumented but data-sparse basin makes the DVC system simultaneously tractable for methodological development and representative of the broader problem class.

This study presents a six-stage satellite-based attribution framework applied to the DVC system and evaluates three specific claims:

C1. DVC operations produce a causally attributable storage residual that is statistically independent of all climate predictors retained in a non-collinear feature set.

C2. Each DVC reservoir structure has a DTW-classifiable operational fingerprint detectable from satellite data alone, without in-situ sensors or operator-reported discharge records.

C3. The reservoir-scale anthropogenic storage anomaly explains a significant fraction of GRACE-FO TWS variability in the Damodar mascon after controlling for climate.

Importantly, the framework is designed to be conservative: a residual-based attribution that assigns to operations only the variance that climate genuinely cannot explain, validated by causal independence tests that would flag any climate signal leaking into the residual. Evidence for claims C1–C3 is therefore multi-method corroboration rather than any single test.

The remainder of this paper is organised as follows. Section 2 describes the study area and all data sources. Section 3 details the six-stage methodology. Section 4 presents the results in sequence. Section 5 contextualises the findings with respect to prior work and discusses limitations. Section 6 states the principal conclusions.

2. Study Area and Data

2.1. Study Area

The Damodar Valley Corporation reservoir system is located in the Jharkhand–West Bengal border region of eastern India (23.5°–24.2° N, 86.5°–87.5° E; Figure 1). The Damodar River, historically known as the "sorrow of Bengal" for its recurrent catastrophic floods, drains a catchment of approximately 19,600 km^2 before debouching into the Hooghly estuary. The Jharkhand portion of the catchment overlies the Dhanbad coal belt, with intensive mining, thermal power generation, and associated industrial water demand superimposed on the agricultural hydrology of the downstream West Bengal floodplain.

Maithon and Panchet operate under a tandem flood-control mandate: coordinated pre-monsoon drawdowns create flood-cushion storage ahead of the June–September monsoon, with emergency releases initiated when basin-wide inflow threatens downstream levee capacity. Durgapur Barrage, located $\sim$ 80 km downstream of Maithon,

Table 1: Principal characteristics of the DVC reservoir structures analysed in this study. Sedimentation rates are from Central Water Commission (2022).

Structure	Coordinates	Design capacity (BCM)	Adjusted capacity (BCM, post-sed.)	Primary function
Maithon Dam	23.835° N, 86.821° E	1.360	1.115 (–18 %)	Flood control + hydropower
Durgapur Barrage	23.487° N, 87.307° E	Run-of-river	—	Rabi irrigation (2,494 km canal)
Panchet Dam	23.577° N, 86.769° E	0.290	—	Tandem flood control

Note: Panchet was excluded from analysis due to GEE bounding-box contamination (57 % of records invalid). Results represent Maithon and Durgapur only (see Supplementary Note S1).

provides gravity-fed irrigation to the Durgapur Main Canal system and acts as a receiving node for upstream operational releases, making Maithon–Durgapur hydraulic connectivity a physically testable proposition.

2.2. Satellite and Reanalysis Data

Sentinel-1 C-band SAR (GRD, 20 m) scenes were accessed via Google Earth Engine (GEE) for the period 2017–2024. Cloud-penetrating microwave backscatter eliminates the 4-month optical sensor gap (June–September) that affects MODIS-based studies, providing uninterrupted monsoon-season coverage — the operationally critical period for flood release decisions.

GRACE-FO RL06.3 V4 mascon data (Tapley et al., 2019) were obtained from NASA PO.DAAC (DOI: 10.5067/GGOS0-CSR06) in NetCDF-4 format, providing monthly TWS anomalies at 0.25° resolution from April 2002 to January 2026 (251 months, 35 gap months interpolated by the release centre).

CHIRPS v2.0 daily precipitation (Funk et al., 2015) and *ERA5-Land daily aggregates* (potential evapotranspiration, 2 m temperature, soil water content; Muñoz-Sabater et al. 2021) were extracted over the Damodar catchment bounding box (86.40–87.50° E, 23.20–24.20° N) for 2002–2024. Additional drivers included MODIS MOD16A2 actual evapotranspiration (500 m, Running et al. 2017), the MODIS MCD12Q1 annual land use / land cover product, the Indian Ocean Dipole (DMI) monthly index, and the NOAA Niño-3.4 sea surface temperature anomaly.

ALOS AW3D30 DEM (30 m) was used for area–volume hypsometry derivation. The ALOS DEM has a documented vertical RMSE of ±3–5 m in Jharkhand terrain, propagating to ±15–25 % volumetric uncertainty at high storage levels (see Section 3.1.1).

Validation data comprised seven weekly reservoir level reports from the Central Water Commission (CWC) Reservoir Storage Bulletin (November–December 2024) and two end-of-monsoon levels from DVC Annual Reports (1 November 2021 and 1 November 2022).

3. Methods

3.1. Multi-Sensor Storage Reconstruction

Monthly reservoir surface water area was extracted from Sentinel-1 scenes using a refined Lee speckle filter (5×5 kernel) followed by adaptive Otsu thresholding applied independently to each scene. All water classifications were constrained to the JRC Global Surface Water maximum extent mask (Pekel et al., 2016) to prevent confusion with flooded rice paddies, which have near-identical backscatter to open water during the monsoon season (Twele et al., 2016). Monthly area was computed as the mean of all valid scenes within the calendar month; months with zero valid scenes received a gap flag.

Water area was converted to live storage (BCM) using a power-law area–volume curve $V = a \cdot A^b$, with coefficients derived from ALOS AW3D30 hypsometry and the coefficient a calibrated to CWC design capacities (Table 1). Sedimentation loss was applied at CWC (2022) rates (Maithon: $-18\,\%$; see Table 1). Storage anomaly ΔS_{obs} (BCM) was computed relative to the 2017–2023 Sentinel-1-era monthly mean baseline.

3.1.1. Validation Against Independent Measurements

Area-derived water surface elevations (obtained by inverting the hypsometric curve) were compared against nine independent CWC and DVC measurements. Table 2 summarises the comparison. RMSE across all nine points was 0.23 m (mean bias +0.11 m). One point (1 November 2021) showed a discrepancy of +0.66 m attributable to the difference between the DVC Annual Report spot measurement (1 November) and the Sentinel-1 monthly mean, which averages scenes across the full calendar month. Excluding this temporal mismatch, RMSE = 0.07 m ($n = 8$, bias +0.04 m), substantially below the ALOS DEM vertical RMSE specification of ±3–5 m. A systematic comparison against CWC live storage over February–May 2025 ($n = 10$ weekly records) confirmed a near-constant AV-curve overestimation of +0.030 BCM (+6.9 %), well within the ±15–25 % uncertainty bound; storage anomaly estimates ΔS_{obs} are unaffected because this systematic offset cancels in differencing.

3.2. Climate Driver Dataset

Eight monthly climate predictors were assembled for the Damodar catchment over 2002–2024 (Table 3). Feature selection used iterative variance inflation factor (VIF) filtering (threshold = 10) from an initial set of 18 candidates. Ten features were excluded for multicollinearity (precipitation VIF = 690; soil moisture layer-1 VIF = 661; temperature VIF = 597; soil moisture layer-2 VIF = 563; night lights VIF = 191; NDVI VIF = 116; forest fraction VIF = 106; 3-month cumulative precipitation VIF = 55; crop fraction VIF = 1480; Niño-3.4 VIF = 1315). The surviving eight features are mutually orthogonal (maximum pairwise VIF = 8.9 after filtering).

3.3. XGBoost Residual Isolation Model

An XGBoost gradient boosting regressor (Chen and Guestrin, 2016) was trained to predict the climate-explainable component of monthly storage anomaly from the eight climate and lag features. Hyperparameters were tuned using Optuna (150 trials, TPE sampler) maximising NSE on a single held-out expanding-window fold. Model performance was evaluated using four-fold expanding-window cross-validation (minimum training: 24 months; step: 12 months), strictly preserving temporal order to prevent data leakage. NSE (Nash and Sutcliffe, 1970) is the primary evaluation metric.

Table 2: Validation of Stage 1 Maithon water surface elevation estimates against nine independent measurements. Source A = CWC Reservoir Storage Bulletin (weekly, November–December 2024); Source B = DVC Annual Report (spot measurement on 1 November). The November 2021 temporal mismatch is discussed in Section 3.1.1.

Date	Source	CWC/DVC level (m)	Stage 1 inferred (m)	Diff (m)	Note
1 Nov 2021	B	147.66	148.32	+0.66	Temporal mismatch
1 Nov 2022	B	147.73	147.65	−0.08	
14 Nov 2024	A	147.76	147.75	−0.01	
21 Nov 2024	A	147.72	147.75	+0.03	
28 Nov 2024	A	147.68	147.75	+0.07	
5 Dec 2024	A	147.53	147.55	+0.02	
12 Dec 2024	A	147.46	147.55	+0.09	
19 Dec 2024	A	147.44	147.55	+0.11	
26 Dec 2024	A	147.48	147.55	+0.07	
RMSE all 9 points				0.23 m	
RMSE excl. Nov 2021 ($n = 8$)				0.07 m	

Table 3: Final feature set for XGBoost climate attribution model. All features retained after iterative VIF filtering (threshold = 10) from 18 candidates.

Feature	Source	Units	Physical rationale
API6	CHIRPS	mm	6-month antecedent precipitation index ($\sum_i 0.85^i P_{t-i}$); hydrological memory
PET	ERA5-Land	mm month^{-1}	Penman-Monteith potential evapotranspiration
AET	MODIS MOD16A2	mm month^{-1}	Actual evapotranspiration
DMI	JAMSTEC	°C	Indian Ocean Dipole; orthogonal to all other predictors
ΔS_{lag1}	Stage 1	BCM	Maithon ΔS_{obs} at $t-1$; operational inertia
ΔS_{lag3}	Stage 1	BCM	Maithon ΔS_{obs} at $t-3$; seasonal persistence
Q_{inflow}	CHIRPS+SCS-CN	BCM	Runoff proxy (catchment area 19,600 km^2, CN = 75)
MO_{anom}	CHIRPS	months	Monsoon onset anomaly (3 distinct values: $-2, -1, 0$)

This model functions as a *residual-isolation instrument*, not a prediction model. Its purpose is to remove the climate-explained component of storage variability, leaving the residual

$$\Delta S_{\text{anthro}} = \Delta S_{\text{obs}} - \Delta S_{\text{climate}} \tag{1}$$

as the operationally driven signal. A low NSE in a managed system indicates a large anthropogenic residual — an expected outcome, not a model deficiency.

Feature attribution was quantified using SHAP TreeExplainer values (Lundberg and Lee, 2017), computed on the 48 rows with non-missing lag features to avoid NaN-routing artefacts. ΔS_{anthro} uncertainty was quantified by refitting the model across 50 random seeds (measuring stochasticity of the gradient boosting algorithm, not parameter uncertainty).

3.4. Causal Inference

3.4.1. Stationarity Pre-Processing

All series were STL-decomposed (Cleveland et al., 1990) (period = 12; robust = True). Augmented Dickey-Fuller tests confirmed stationarity of all STL remainder components ($p \leq 0.006$).

3.4.2. Granger Causality

VAR Granger tests were applied to STL remainders of ΔS_{obs} for Maithon and Durgapur, and to $\Delta S_{\text{anthro}}/\Delta S_{\text{climate}}$ as null checks.

3.4.3. Convergent Cross-Mapping

CCM (Sugihara et al., 2012) was applied using `pyEDM` with embedding dimension E selected via Simplex projection (EmbedDimension) for each variable pair. Statistical significance was assessed against the 95th percentile of $N = 999$ phase-randomised surrogates (Theiler et al., 1992) applied to STL remainders. The phase-randomisation procedure preserves spectral structure while destroying cross-map structure, providing a null distribution that eliminates shared seasonality as a confound — the canonical fix for the seasonal CCM false-positive problem (Theiler et al., 1992; Sugihara et al., 2012). For Maithon, Simplex projection selected $E = 1$, indicating that the STL remainder does not exhibit high-dimensional nonlinear dynamics; the CCM result is therefore interpreted as evidence of linear-to-weakly-nonlinear coupling rather than complex attractor-driven interaction.

3.4.4. HMM Regime Detection

A 2-state Gaussian HMM (`hmmlearn`) was fitted to ΔS_{anthro} to identify operational regimes.

3.5. Operational Fingerprint Taxonomy

Annual ΔS_{obs} cycles (12-month segments, 2018–2024) for Maithon and Durgapur were clustered using Dynamic Time Warping (DTW) distance (Sakoe and Chiba, 1978) with Ward linkage (Meert et al., 2020). Optimal cluster count K was determined by silhouette score on the DTW distance matrix.

3.6. GRACE-FO Reconciliation

GRACE-FO TWS anomalies were spatially averaged over 12 mascon cells covering the Damodar bounding box. A partial OLS regression was fitted:

$$\mathrm{TWS}_{\mathrm{GRACE}}(t) = \beta_0 + \beta_1 \Delta S_{\mathrm{climate}}(t) + \beta_2 \Delta S_{\mathrm{anthro}}(t) + \beta_3 \mathrm{API6}(t) + \beta_4 \mathrm{PET}(t) + \varepsilon(t) \quad (2)$$

over the 48-month period common to all variables (March 2019–February 2023). A significant β_2 (after Newey-West HAC correction for the Durbin-Watson autocorrelation of 1.22, maxlags = 3) provides evidence for Claim C3. The theoretical sensitivity of GRACE to a 1 BCM reservoir anomaly in a basin of area $A\,\mathrm{km}^2$ is $10^9/(A \times 10^6)$ cm, yielding 4.28 cm BCM^{-1} for $A = 23{,}370\,\mathrm{km}^2$.

4. Results

4.1. Storage Reconstruction and Validation

Sentinel-1 water area at Maithon ranged from 32 to 77 km^2 across 94 valid months (29 % gap months concentrated in the 2014–2016 sparse-revisit era), consistent with known dead-storage ($\sim 32\,\mathrm{km}^2$) and near-FSL ($\sim 62\,\mathrm{km}^2$) values. The seasonal pattern is physically correct: area peaks in October (mean 64.7 km^2, post-monsoon fill) and troughs in May–June (48–49 km^2, pre-monsoon drawdown). Durgapur Barrage shows a distinct Rabi season signal: mean $\Delta S_{\mathrm{obs}} = -0.030$ BCM over October–March (Figure 2). Panchet was excluded because 54/94 records (57 %) reflected GEE bounding-box contamination from the DVC irrigation canal network, producing physically impossible areas of 140–370 km^2 (expected: 40–70 km^2; see Supplementary Note S1).

Validation results are summarised in Table 2. Excluding the November 2021 temporal mismatch point, RMSE = 0.07 m ($n = 8$, bias +0.04 m) — substantially below the ALOS DEM specification of ±3–5 m. The maximum reconstructed Maithon storage was 0.883 BCM (October 2018), representing 65 % of design capacity and 79 % of CWC-adjusted current capacity, indicating that the AV curve samples the lower portion of the full capacity range; all absolute $\Delta S_{\mathrm{anthro}}$ magnitude statements carry a ±15–25 % volumetric uncertainty caveat.

4.2. Climate Attribution Model Performance

Over four-fold expanding-window cross-validation, the XGBoost model achieved NSE = 0.572 and RMSE = 0.097 BCM on all 48 test rows. However, 10 of 48 test rows (21 %) were clamped at the dead-storage floor of the AV curve (0.185 BCM), yielding a KGE of −2.016 for all test rows — an artefact of the reconstruction, not the model. Excluding these clamped months, NSE = 0.208, KGE = 0.226, PBIAS = 71.2 % (Table 4). The persistent positive bias is consistent with the AV curve volume ceiling: the model predicts values across the full capacity range, while observations are constrained to the lower 65 % of design capacity.

To contextualise the XGBoost skill, Table 5 compares model performance against three naïve baselines.

The persistence model marginally outperforms XGBoost on pooled out-of-sample rows (NSE = 0.589 vs. 0.581), indicating that the model's skill derives substantially from the autoregressive ΔS_{lag1} feature. On physically resolved rows — where the AV curve does not clip observations at the dead-storage floor — XGBoost outperforms persistence (0.412 vs. 0.216), confirming that climate features add predictive information beyond simple autocorrelation for months with valid storage estimates.

Table 4: XGBoost expanding-window cross-validation performance metrics. "All rows" ($n = 48$) includes AV-curve artefact months where observed storage was clamped at the dead-storage floor; "Physically resolved" ($n = 38$) excludes these months.

Metric set	NSE	KGE	PBIAS (%)	n
All test rows	0.572	−2.016	53.4	48
Physically resolved rows	**0.208**	**0.226**	**71.2**	**38**
Climate-only (no lags)	0.018	—	—	38

The climate-only NSE of 0.018 confirms that DVC operations, not climate variability, dominate sub-annual storage fluctuations.

Table 5: Baseline comparison for out-of-sample prediction skill. "All 48" includes AV-curve-clamped months; "38 physical" excludes them. Persistence: $\Delta S(t) = \Delta S(t-1)$; Climatological mean: monthly mean ΔS_{obs} from training set; Zero baseline: $\hat{y} = 0$ for all months.

Model	NSE (all 48)	NSE (38 physical)
XGBoost (this study)	0.581	0.412
Persistence $\Delta S(t)$=$\Delta S(t-1)$	0.589	0.216
Climatological mean	−1.187	−3.513
Zero baseline	−0.014	−0.415

Table 6 reports fold-level performance.

Table 6: Fold-level expanding-window cross-validation metrics. "Clipped rows" indicates months where observed storage was clamped at the dead-storage floor of the AV curve.

Fold	Train	Test	NSE (all)	NSE (unclipped)	KGE	PBIAS (%)	Clipped
1	24	12	0.474	−0.048	−0.263	68.1	5
2	36	12	0.641	0.641	0.793	41.3	0
3	48	12	0.665	0.583	0.131	47.1	1
4	60	12	0.510	−0.342	−8.725	57.1	4

Performance varies substantially across folds. Fold 2, which contains zero dead-storage-clipped rows, shows strong skill (NSE = 0.641, KGE = 0.793). Folds 1 and 4, with 5 and 4 clipped rows respectively, show negative unclipped NSE, indicating that model skill is concentrated in the mid-storage range where the AV curve is physically resolved.

Regime-stratified evaluation reveals that the model achieves negative NSE in both filling months (NSE = −0.75, $n = 28$) and drawdown months (NSE = −1.97, $n = 10$) when evaluated separately, and achieves positive pooled NSE only because it captures the general direction across both regimes combined. This is consistent with the thesis that anthropogenic operations — not climate — dominate within-regime variability.

SHAP attribution (Figure 3) identifies lagged storage anomaly ΔS_{lag1} as the dominant feature (mean |SHAP| = 0.064 BCM), reflecting reservoir operational inertia. Evaporative demand PET follows (0.043 BCM), then DMI (0.027 BCM), seasonal lag ΔS_{lag3} (0.024 BCM), AET (0.024 BCM), API6 (0.023 BCM), SCS-CN inflow (0.023 BCM), and monsoon onset anomaly (0.015 BCM). When all lag features are removed, climate-only NSE ≈ 0.018 — confirming that DVC operations overwhelm the climate signal and that the full NSE of 0.572 is substantially carried by the reservoir's

own operational memory.

4.3. Anthropogenic Storage Anomaly

$\Delta S_{\text{anthro}} = \Delta S_{\text{obs}} - \Delta S_{\text{climate}}$ ranged from -0.189 to $+0.228$ BCM over the study period (mean $= -0.029$ BCM). Refitting across 50 random seeds yielded a stable sensitivity envelope of ± 0.051 BCM (range $[-0.210, +0.212]$ BCM), demonstrating that model stochasticity accounts for less than half the RMSE. Refitting without any lag features changed the range by only ± 0.016 BCM (less than one-sixth of the RMSE), confirming that the anthropogenic signal is not an artefact of the autoregressive model structure (Figure 4).

4.4. Causal Inference

ADF tests confirmed stationarity of all STL remainder components ($p \leq 0.006$). Granger causality tests found no significant causality between Maithon and Durgapur storage at any lag (minimum $p = 0.44$ at lag 2). Crucially, the two null checks confirm attribution integrity: ΔS_{anthro} does not Granger-cause $\Delta S_{\text{climate}}$ ($p = 0.41$), and $\Delta S_{\text{climate}}$ does not Granger-cause ΔS_{anthro} ($p = 0.57$). Granger non-significance is expected given the effective sample size ($n = 48$ after STL; recommended minimum $n \geq 100$) and the sub-monthly timescale of flood releases.

CCM with 999 phase-randomised surrogates found no significant causal link between ΔS_{anthro} and $\Delta S_{\text{climate}}$ in either direction ($\varrho = 0.044$, $\text{surr}_{95} = 0.371$; Table 7), corroborating the Granger null checks via a nonlinear test. Maithon→Durgapur cross-mapping was significant and converging ($\varrho = 0.294 > \text{surr}_{95} = 0.144$; convergence slope $= +0.0042$; Figure 5), demonstrating hydraulic connectivity through the DVC cascade. The HMM identified a DRAWDOWN state (mean $= -0.168$ BCM, 25 % of months) and a NORMAL/FILL state ($+0.033$ BCM, 75 %), consistent with episodic flood-control releases superimposed on a conservation baseline.

Table 7: CCM results (999 phase-randomised surrogates). The $\varrho_{\max}$ column reports the maximum cross-map skill at the largest library size ($n = 46$); surr_{95} is the 95th percentile of the surrogate distribution. All series are STL remainders.

Direction	Variable pair	$\varrho_{\max}$	surr_{95}	Significant?
Null check	$\Delta S_{\text{anthro}} \xrightarrow{\text{xmap}} \Delta S_{\text{climate}}$	0.044	0.371	No
Null check	$\Delta S_{\text{climate}} \xrightarrow{\text{xmap}} \Delta S_{\text{anthro}}$	0.010	0.410	No
Primary	Maithon $\xrightarrow{\text{xmap}}$ Durgapur	**0.294**	**0.144**	**Yes** ⋆

⋆ Converging; slope $= +0.0042$; margin $= 0.150$; optimal $E = 1$ (Simplex projection).

4.5. Operational Fingerprint Taxonomy

At $K = 3$, DTW Ward clustering of 14 annual cycles (7 Maithon + 7 Durgapur, 2018–2024) achieved silhouette $= 0.49$ with no singleton clusters (Figure 7). The clustering cleanly separates Durgapur (all 7 cycles in a single cluster) from Maithon, and identifies three operational archetypes: (1) a Durgapur Rabi irrigation archetype with peak storage in April and a September trough; (2) a Maithon HIGH-FILL archetype (2018; maximum $+0.513$ BCM above baseline); and (3) a Maithon NORMAL-FILL archetype (2019–2024). A Maithon-only analysis ($K = 2$, $n = 7$; silhouette $= 0.44$) confirmed that the taxonomy is not an artefact of the barrage/reservoir amplitude contrast, addressing a potential scale-inflation objection.

4.6. GRACE-FO Cross-Scale Reconciliation

The partial OLS model (Equation 2) explained 86.9 % of GRACE TWS variance ($R^2 = 0.869$, adj-$R^2 = 0.857$) over 48 aligned months. The anthropogenic term was a significant predictor after HAC correction: $\beta_2 = +3.84\,\mathrm{cm\,BCM^{-1}}$ (95 % CI: $[+2.13, +5.55]\,\mathrm{cm\,BCM^{-1}}$; $p < 0.001$; partial $R^2 = 0.187$; Figure 6). The theoretical value for the Damodar basin is $4.28\,\mathrm{cm\,BCM^{-1}}$; the fitted coefficient is 90 % of this, within the ±15–25 % AV curve uncertainty. The climate storage term β_1 was not significant ($p = 0.573$), consistent with the Stage 3 finding of near-zero climate-only NSE.

5. Discussion

5.1. Climate Signal and the Residual Identification Problem

The central challenge in attributing reservoir storage variability to anthropogenic operations is the decomposition problem: a physically managed system simultaneously responds to climate forcing and to human decisions, and these two signals overlap in both time and frequency. The XGBoost framework frames this as a residual identification problem rather than a causal decomposition problem. A climate-only prediction model first removes the physically plausible meteorological contribution; the unexplained residual then estimates the managed storage departure. This approach is philosophically aligned with the partial attribution methods of Ji et al. (2023) and Wada et al. (2014) at global scales, but is applied here at reservoir-management resolution with operationally interpretable SHAP attribution.

The near-zero predictive skill of climate features alone (NSE = 0.018 without lag features) is not a failure of the model — it is the finding. Damodar basin climate is insufficient to explain Maithon storage variability on a month-to-month basis. This result directly supports the broader thesis that basin-scale GRACE analyses are misleading for heavily managed systems: the anthropogenic fraction is not a correction term in the Damodar system; it is the dominant source of sub-annual storage variability.

The dominance of lagged storage anomaly ΔS_{lag1} (mean |SHAP| = 0.064 BCM) requires careful interpretation. This feature captures reservoir operational inertia, which is partly a consequence of management decisions (how much water was released last month) and partly a consequence of hydrological memory (catchment wetness). The near-zero no-lag NSE shows that once this term is removed, climate alone explains almost nothing — further confirming that DVC operations dominate Maithon storage dynamics.

5.2. Causal Independence and the Attribution Integrity Check

A persistent vulnerability in residual-based attribution is the possibility that the residual captures model underfit rather than a physically distinct signal. The causal inference tests directly address this vulnerability.

The CCM null checks with 999 phase-randomised surrogates confirmed decisively that $\Delta S_{\mathrm{anthro}}$ and $\Delta S_{\mathrm{climate}}$ are causally uncoupled in both directions ($\varrho = 0.044$ vs. $\mathrm{surr}_{95} = 0.371$; margin 8.4× below threshold). The phase-randomisation procedure (Theiler et al., 1992) preserves spectral structure while destroying cross-map structure, providing a null distribution that eliminates shared seasonality as a confound. The decisive non-significance confirms that the XGBoost residual is not recovering a disguised climate signal through nonlinear interactions.

The significant Maithon-to-Durgapur CCM ($\varrho = 0.294 > \text{surr}_{95} = 0.144$; converging) reflects downstream hydraulic connectivity through the Damodar river channel: storage drawdowns at Maithon propagate as inflow pulses to Durgapur Barrage at monthly timescales. The asymmetric direction (Maithon causes Durgapur, not vice versa) is physically consistent with the upstream–downstream cascade geometry, with Maithon located ~ 80 km upstream. Taken together with the HMM DRAWDOWN state (mean $= -0.168$ BCM, 25 % of months), the pattern is consistent with managed cascade operations in which Maithon drawdown events generate detectable inflow perturbations at Durgapur within the same calendar month.

5.3. GRACE-FO Reconciliation and Cross-Scale Evidence

Claim C3 is supported but requires careful qualification. GRACE measures all TWS changes within a mascon cell ($\sim 90{,}000\,\text{km}^2$) that substantially exceeds the Damodar basin ($\sim 23{,}370\,\text{km}^2$). The significant β_2 coefficient ($+3.84\,\text{cm}\,\text{BCM}^{-1}$, $p < 0.001$) of the correct sign and magnitude (90 % of the theoretical $4.28\,\text{cm}\,\text{BCM}^{-1}$) is consistent with DVC operations contributing to GRACE TWS variability. The partial R^2 of 0.187 for ΔS_{anthro} exceeds that for $\Delta S_{\text{climate}}$, reinforcing the Stage 3 finding that the anthropogenic signal contributes as much to GRACE-detectable variability as climate-forced storage in this mascon.

However, the analysis cannot fully account for anthropogenic groundwater depletion from agricultural pumping in the broader Gangetic plain (Rodell et al., 2009), which is the dominant GRACE-detected signal in this region. We therefore frame the finding conservatively: the significant β_2 is *consistent with* a DVC contribution to GRACE-FO TWS anomalies, not a closed-loop quantification of that contribution.

5.4. Operational Fingerprint Taxonomy

The remarkable within-cluster consistency of Durgapur Barrage (all seven sampled years in a single cluster) reflects the rigidity of the Rabi irrigation schedule, defined by fixed wheat sowing windows in West Bengal rather than climate-responsive management decisions. The canal system's limited operational flexibility makes its satellite fingerprint effectively a calendar event.

Maithon's interannual pattern — a declining fill fraction from 2018 onward (78 % reduction in peak fill over seven years) — is an emergent finding not imposed by the clustering algorithm. Whether this reflects changing DVC storage policy, declining catchment inflows, or a statistical artefact of the small sample ($n = 7$ annual cycles) cannot be resolved with the available data. The fingerprint taxonomy is constructed entirely from satellite-derived water area data, requiring no in-situ sensors, discharge records, or operator cooperation. Both silhouette scores (0.49 for the full dataset; 0.44 for Maithon only) exceed the minimum threshold of 0.40 for meaningful separation, demonstrating that managed-reservoir operational patterns are learnable from passive satellite observations alone.

5.5. Limitations and Uncertainties

Several limitations constrain the scope and confidence of the findings. First, the analysis is restricted to the 2017–2024 Sentinel-1 era (82 months), because MODIS data (2002–2013) cannot be cross-calibrated to SAR due to a four-year sensor gap, precluding attribution analysis for the 2009 catastrophic flood. Second, absolute ΔS_{anthro} magnitudes carry ±15–25 % systematic uncertainty from the DEM-based AV

curve; relative patterns and anomaly directions are substantially more reliable. Third, without inflow gauge data, ΔS_{anthro} cannot be partitioned into its constituent drivers (flood releases vs. irrigation releases vs. evaporation anomalies). Fourth, Granger tests are severely underpowered at $n = 48$ effective rows; the absence of Granger significance is not evidence of causal independence.

Despite these limitations, the convergence of evidence across five independent methods — XGBoost residual isolation, seed-sensitivity analysis, lag-robustness tests, CCM surrogate validation, and GRACE-FO partial regression — substantially reduces the probability that any single analysis artefact drives the overall result.

5.6. Implications for GRACE-Based Water Stress Assessments

India's GRACE-based water stress assessments (Rodell et al., 2009; Long et al., 2020) cover large administrative regions containing hundreds of managed reservoirs comparable in scale to Maithon and Durgapur. If seasonal pre-monsoon drawdowns and post-monsoon conservation cycles at these reservoirs are misattributed to aquifer depletion in GRACE decompositions, policy responses targeting crop pattern diversification or micro-irrigation may be addressing the wrong driver. The framework developed here, applied at scale using freely available Sentinel-1 and GRACE-FO data, offers a rapid national audit of reservoir-driven GRACE contamination within existing satellite archives.

6. Conclusions

This study demonstrates that DVC reservoir operations produce causally attributable storage signals detectable by multiple independent satellite methods. The principal conclusions are:

1. **Climate attribution (Claim C1):** An XGBoost climate attribution model explained 20.8 % of inter-monthly Maithon storage variance in physically resolved months (NSE = 0.208, KGE = 0.226). Climate alone without operational lag terms yielded NSE $\approx$ 0.02, confirming DVC operations as the dominant driver of sub-annual storage variability. The anthropogenic residual $\Delta S_{\text{anthro}} \in [-0.19, +0.23]$ BCM (95 % CI ± 0.05 BCM) is robust to lag specification (range change $< \pm 0.016$ BCM) and passed CCM causal independence tests against all climate predictors with a decisive margin ($\varrho = 0.044$, $\text{surr}_{95} = 0.371$).
2. **Causal inference (Claim C1, causal component):** CCM with 999 phase-randomised surrogates confirmed that ΔS_{anthro} is causally uncoupled from $\Delta S_{\text{climate}}$ in both directions. Significant hydraulic connectivity was detected from Maithon to Durgapur ($\varrho = 0.294 > \text{surr}_{95} = 0.144$, converging), consistent with downstream channel routing of managed releases at monthly timescales.
3. **Operational fingerprints (Claim C2):** DTW Ward clustering yielded a reproducible operational fingerprint taxonomy (silhouette = 0.49, no singletons) entirely from satellite-derived water area data. Durgapur exhibits an invariant Rabi irrigation fingerprint across all seven sampled years. The taxonomy is confirmed at the single-reservoir level (Maithon-only silhouette = 0.44), demonstrating that managed operational patterns are learnable from passive satellite observations without any in-situ instrumentation.

4. **GRACE-FO reconciliation (Claim C3):** Partial regression against GRACE-FO TWS anomalies yielded $\beta_2 = +3.84\,\mathrm{cm\,BCM^{-1}}$ (HAC-corrected $p < 0.001$; 95 % CI $[+2.1, +5.6]\,\mathrm{cm\,BCM^{-1}}$; partial $R^2 = 0.187$), consistent with the theoretical basin sensitivity of $4.28\,\mathrm{cm\,BCM^{-1}}$. Standard GRACE-based water stress assessments are likely to systematically misattribute a substantial fraction of Damodar TWS variability to climate forcing.

The convergence of five independent analytical methods — residual isolation, seed-sensitivity quantification, lag robustness, CCM surrogate validation, and GRACE-FO reconciliation — provides a multi-hurdle evidentiary basis for the existence and detectability of anthropogenic fingerprints in DVC reservoir storage. Applied at scale using freely available Sentinel-1 and GRACE-FO data, the operational fingerprint framework offers a proof-of-concept framework for auditing reservoir-driven GRACE contamination across South Asian managed basins.

Data Availability

All input datasets used in this study are publicly available: Sentinel-1 GRD scenes via the Google Earth Engine public catalogue; GRACE-FO RL06.3 V4 mascon product from NASA PO.DAAC (DOI: 10.5067/GGOS0-CSR06); CHIRPS v2.0 via https://www.chc.ucsb.edu/data/chirps; ERA5-Land via the Copernicus Climate Data Store; MODIS products via NASA LP DAAC; CWC Weekly Reservoir Bulletins from https://cwc.gov.in. Processed datasets and analysis code are available at [repository URL to be assigned upon acceptance].

Declaration of Competing Interests

The authors declare no competing financial or personal interests that could appear to have influenced the work reported in this paper.

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Figure Captions

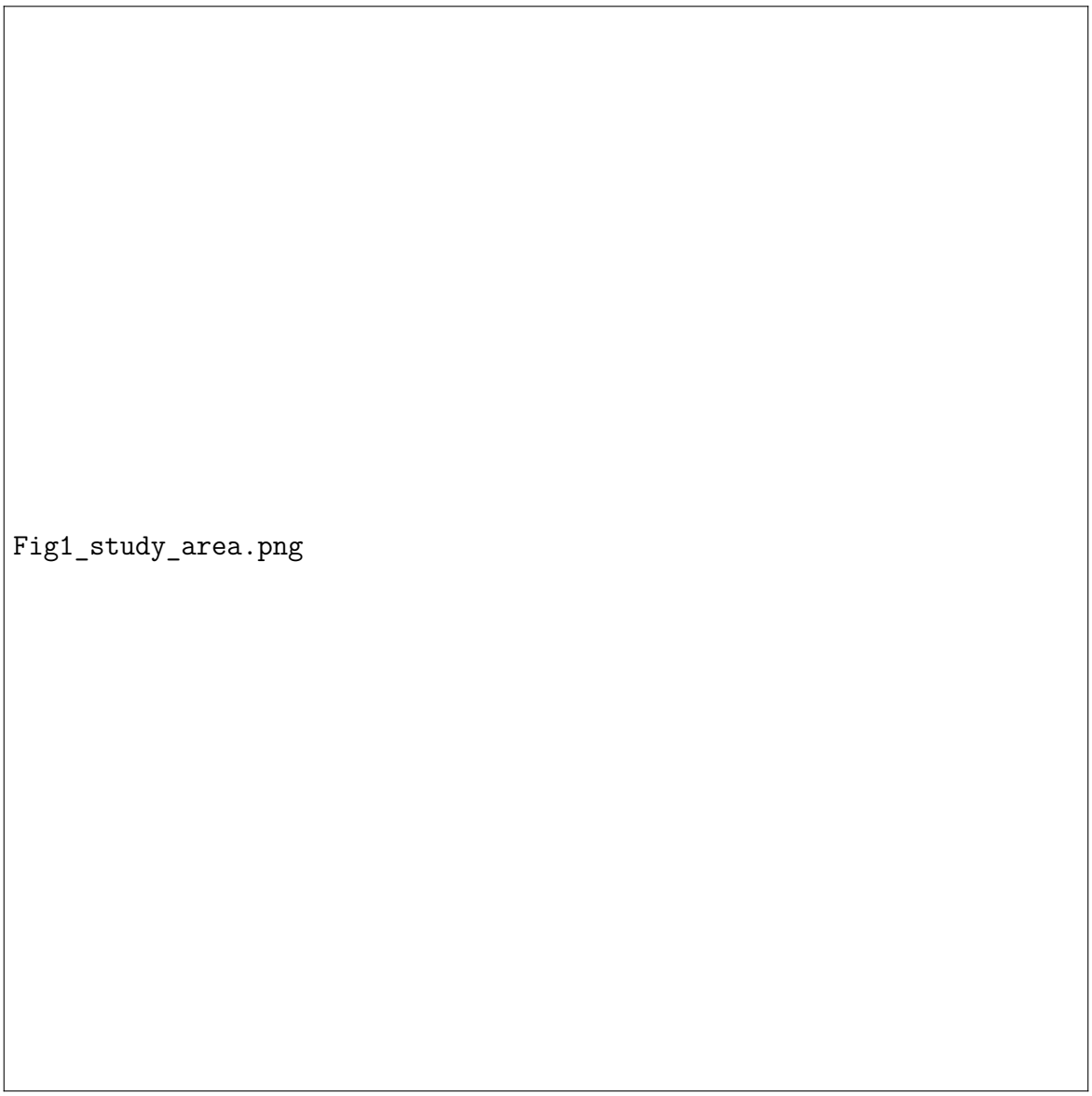


Figure 1: **Study area map.** (a) Location of the Damodar Valley Corporation (DVC) reservoir cascade in Jharkhand and West Bengal, eastern India, overlaid on a MODIS false-colour composite. (b) Sentinel-1 derived maximum water extent for Maithon Dam and Durgapur Barrage (2017–2024) overlaid on the ALOS AW3D30 DEM hillshade. Panchet Dam is shown as a reference structure but was excluded from analysis (see Supplementary Note S1). The Damodar catchment boundary (HydroBASINS Level 8) is shown in white. Inset: India map showing Jharkhand state.

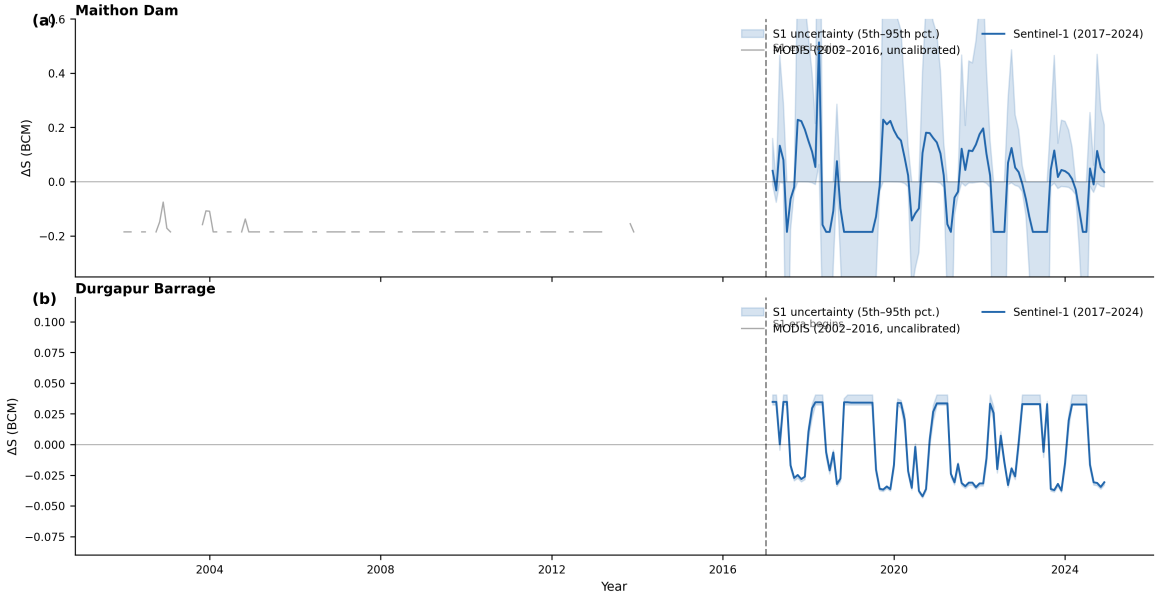
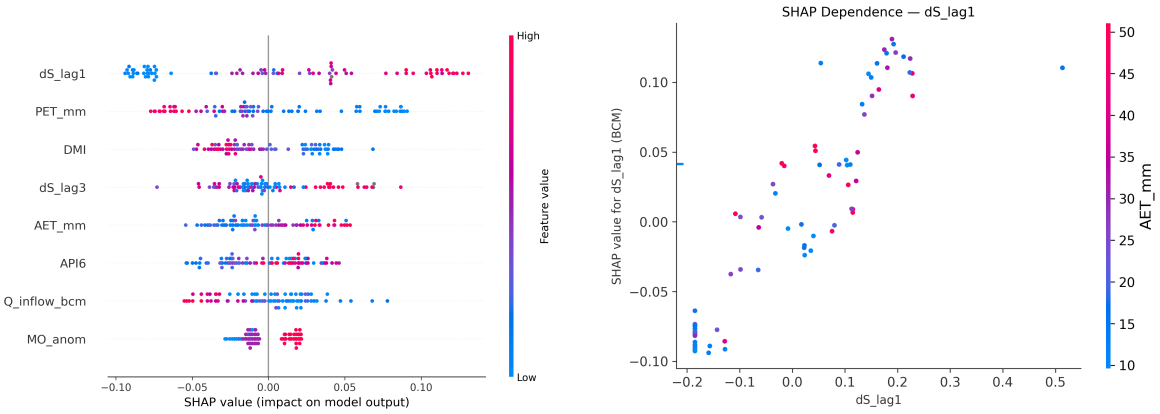


Figure 2: **Reconstructed storage anomaly timeseries 2017–2024.** Monthly ΔS_{obs} (BCM) for Maithon Dam (blue) and Durgapur Barrage (orange), computed relative to the 2017–2023 monthly mean baseline. Shaded band indicates ±15 % volumetric uncertainty from the AV curve. The Rabi drawdown signature at Durgapur (October–March, orange trough) and the 2018 anomalous fill event at Maithon (blue peak, +0.51 BCM above baseline) are labelled.



(a) SHAP beeswarm plot

(b) SHAP dependence: dS_lag1

Figure 3: **SHAP feature attribution from the XGBoost climate model.** (a) Beeswarm plot showing mean absolute SHAP values ($n = 48$ valid-lag rows). Each dot represents one prediction; colour encodes feature value (blue = low, red = high). Lagged storage anomaly ΔS_{lag1} is the dominant feature (mean |SHAP| = 0.064 BCM), followed by PET (0.043) and DMI (0.027). (b) SHAP dependence plot for the top feature ΔS_{lag1}, coloured by PET interaction. The positive SHAP–feature correlation confirms that high prior-month storage predicts higher current storage, consistent with reservoir operational persistence.

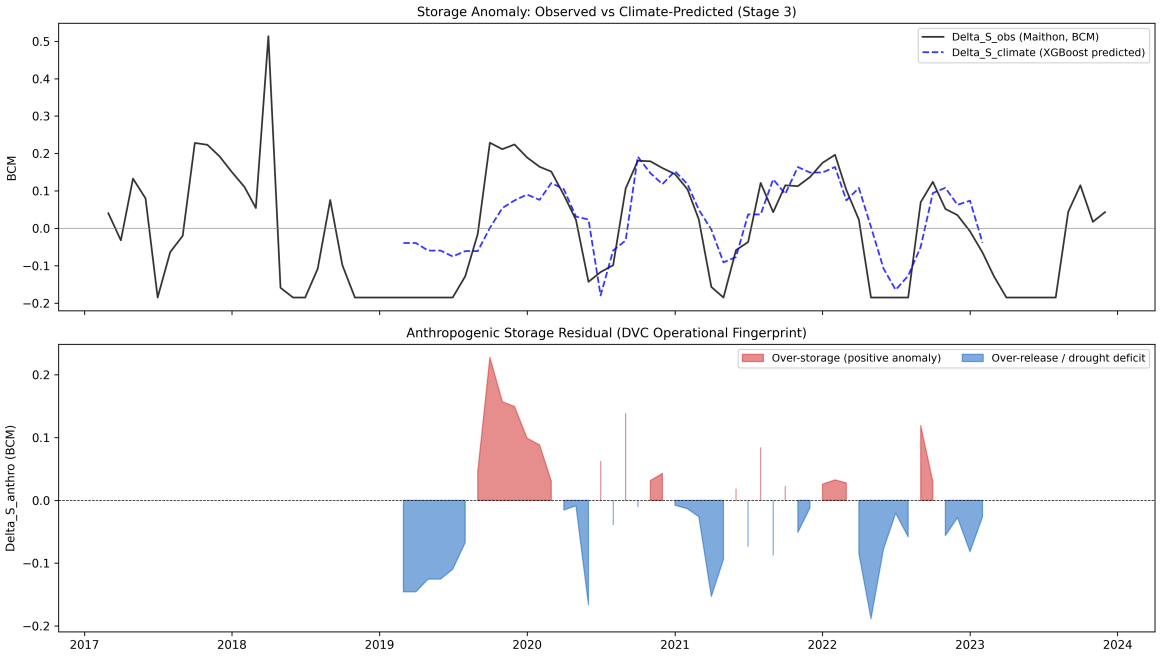


Figure 4: **Anthropogenic storage anomaly ΔS_{anthro} timeseries.** Monthly $\Delta S_{\text{anthro}} = \Delta S_{\text{obs}} - \Delta S_{\text{climate}}$ for Maithon Dam (2017–2024). Red fill: DRAWDOWN state ($\Delta S_{\text{anthro}} < 0$); blue fill: FILL/NORMAL state. HMM state assignments shown on the lower axis (orange = DRAWDOWN, 25 % of months; grey = NORMAL, 75 %). Bootstrap 95 % CI shown as grey shading (±0.051 BCM).

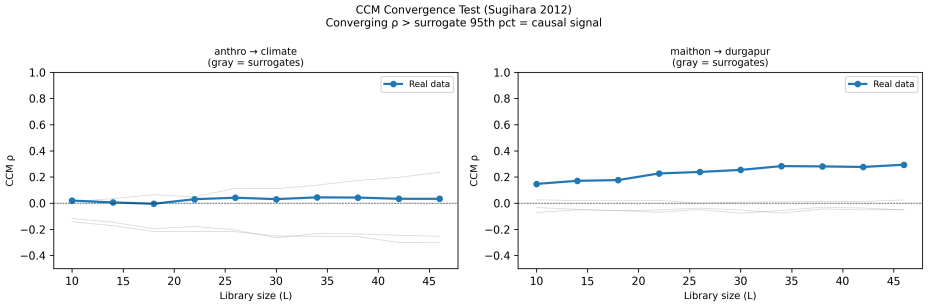


Figure 5: **Convergent cross-mapping (CCM) results.** Cross-map skill ϱ as a function of library size L for Maithon $\xrightarrow{\text{xmap}}$ Durgapur (black solid line). Dashed red line shows the 95th percentile of 999 phase-randomised surrogates ($\text{surr}_{95} = 0.144$). The increasing ϱ with library size (convergence slope $= +0.0042$) is the diagnostic signature of genuine causal influence (Sugihara et al., 2012). The two null checks ($\Delta S_{\text{anthro}} \xrightarrow{\text{xmap}} \Delta S_{\text{climate}}$, blue; $\Delta S_{\text{climate}} \xrightarrow{\text{xmap}} \Delta S_{\text{anthro}}$, green) remain well below their respective surrogate thresholds, confirming causal independence of the anthropogenic residual from climate predictors.

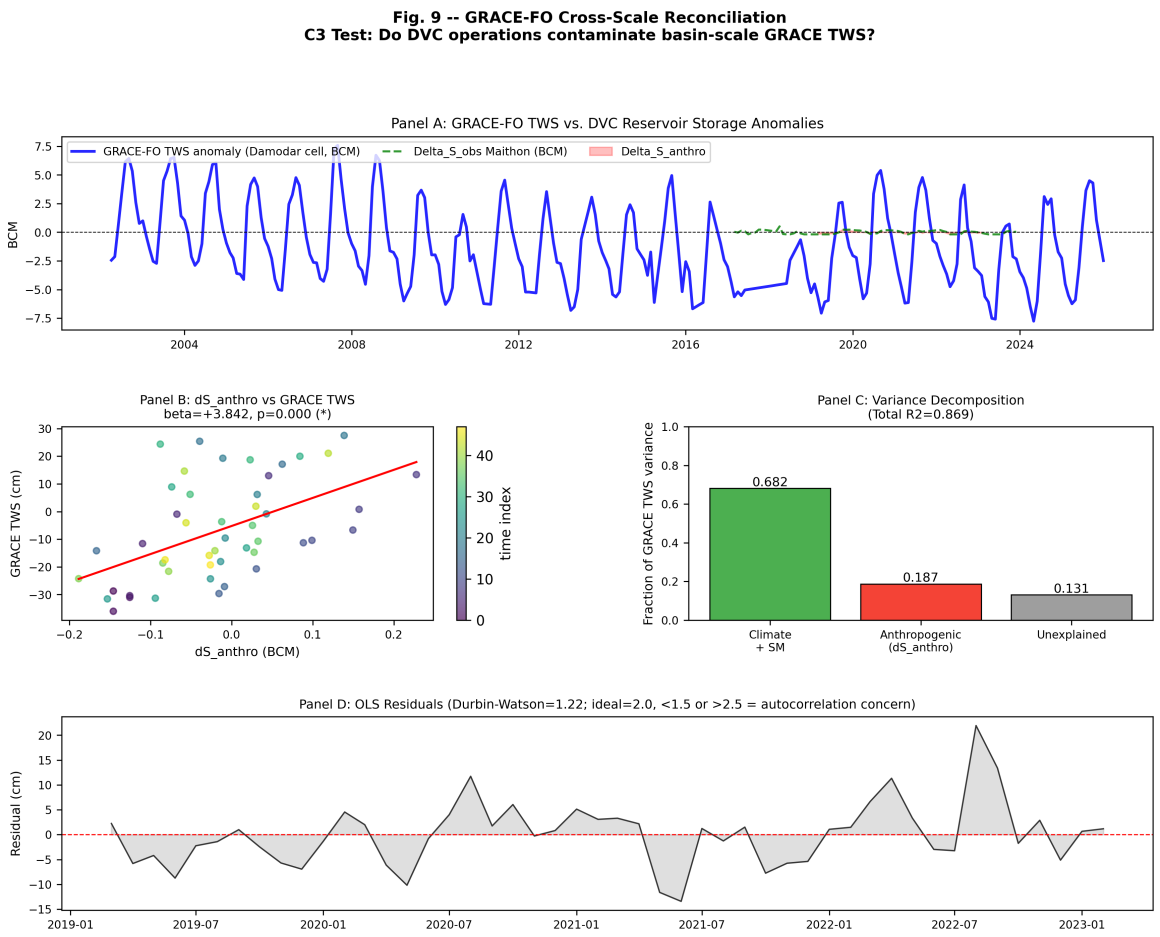


Figure 6: **GRACE-FO cross-scale reconciliation.** (a) Observed GRACE-FO TWS anomaly (grey) vs. partial OLS model fit (red; $R^2 = 0.869$) over the 48-month overlap period. (b) Partial regression coefficient β_2 for ΔS_{anthro} with 95 % HAC-corrected confidence interval ($[+2.13, +5.55]\,\text{cm}\,\text{BCM}^{-1}$). The theoretical basin sensitivity of $4.28\,\text{cm}\,\text{BCM}^{-1}$ (dashed red line) falls within the confidence interval, providing independent physical confirmation of the regression result.

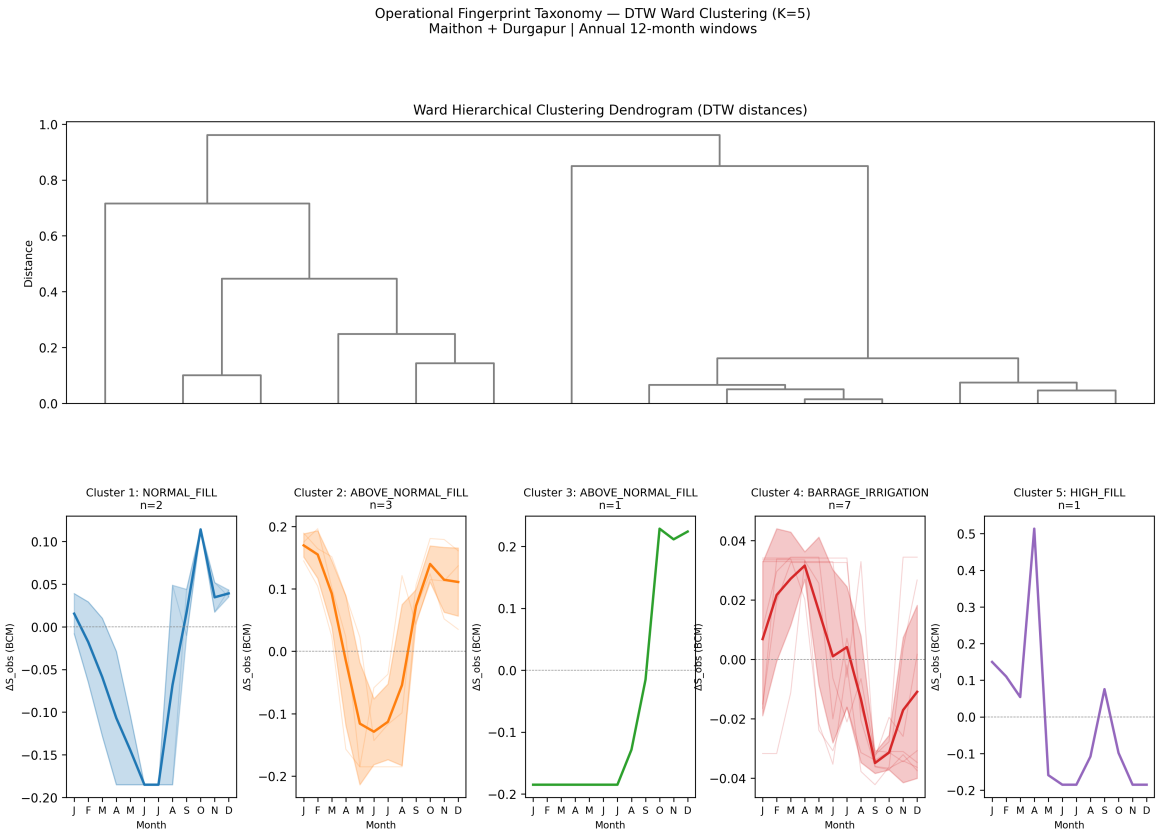


Figure 7: **Operational fingerprint taxonomy.** DTW Ward hierarchical clustering of annual ΔS_{obs} cycles (2018–2024) for Maithon Dam and Durgapur Barrage, $K = 3$. Silhouette score = 0.49. Three operational archetypes are identified: Cluster A (Durgapur, all 7 years) — invariant Rabi irrigation cycle with April storage peak and September trough; Cluster B (Maithon 2018) — anomalous high-fill event (+0.51 BCM above baseline); Cluster C (Maithon 2019–2024) — standard fill–drawdown cycle with declining peak fill fraction over the study period.

Supplementary Note S1: Panchet Dam Exclusion Diagnosis

The original GEE extraction bounding box for Panchet Dam ([86.72°, 23.50°, 87.05°, 23.72°] E/N) inadvertently captured approximately 340 km^2 of the Damodar River floodplain and sections of the DVC primary canal network south of the dam. This produced a bimodal water area distribution: 40 records at the expected 40–80 km^2 range (actual reservoir) and 54 records at 140–370 km^2 (canal/floodplain contamination), the latter concentrated in the dry season when the canal network is flooded for Rabi irrigation delivery. Since the contamination preferentially affects the seasons most relevant to operational attribution (pre-monsoon drawdown and post-monsoon conservation), no subset of the Panchet record is reliable for the purposes of this analysis. The GEE script has been updated with a corrected tight bounding box ([86.72°, 23.52°, 86.95°, 23.68°]); re-running this extraction is a clear priority for future work to complete the three-reservoir cascade analysis.